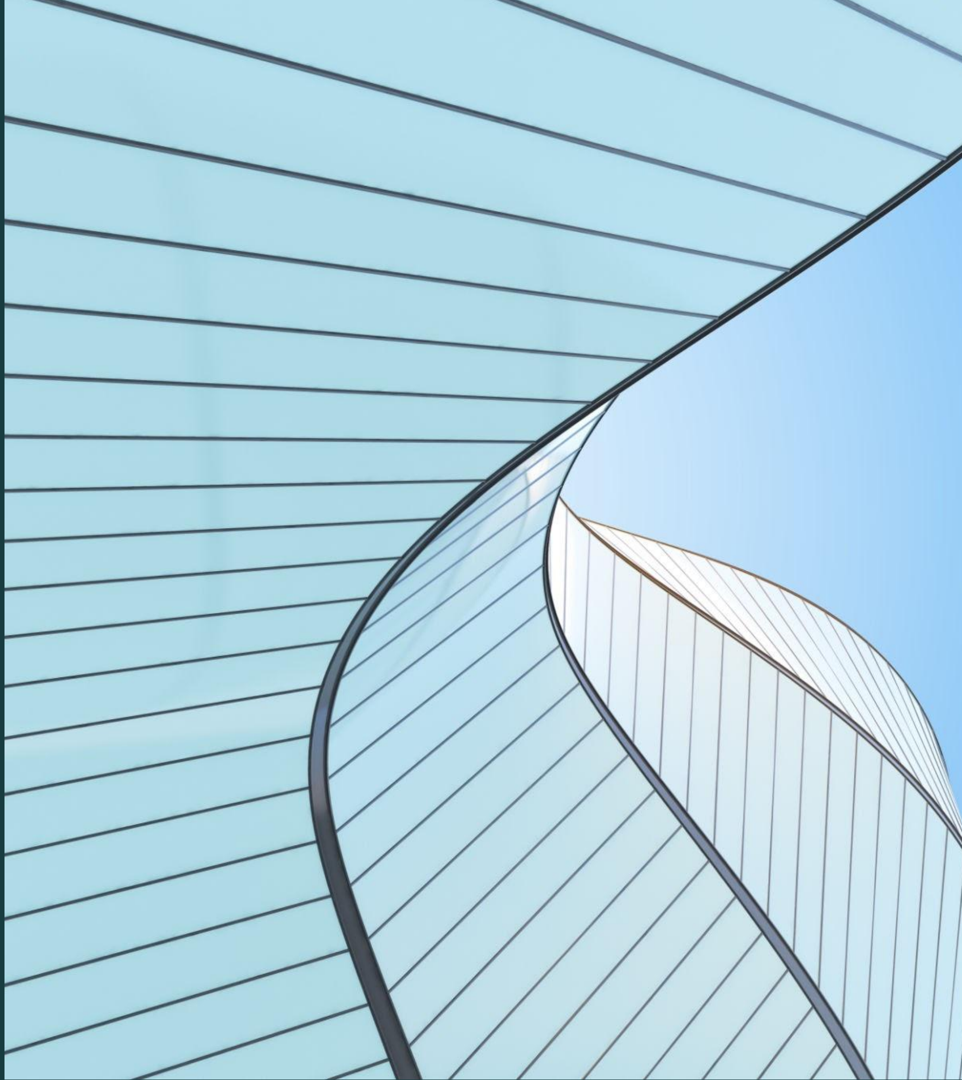
PSTAT 100 Final Project

By Cyrus Isaacs, Tai Ryan, and Bailey Gutierrez



Topic of interest and Research question:

Which MLB games provide the best balance of affordability and fan experience for the 9/9/9 challenge?

Motivation:

We are MLB fans who want to enjoy entertaining baseball games at reasonable costs. As fans we want to find the lowest cost of these, while still watching an entertaining game in order to get the most out of our money spent.

Context:

The 9/9/9 challenge is consuming 9 beers, 9 hotdogs, in 9 innings, one of each per inning.

Goal:

Create a normalized index with mean of 100 and standard deviation of 10, to find the best games in the context of the 9/9/9 challenge as well as predict the optimality of future games. The index will take into account the price of concessions, the odds of the home team winning, and whether any 'superstars' are scheduled to play.

Methods

As a result, game enjoyability, team performance, and making an optimized guess to predict the best experience you can get from a MLB game

An entertaining game consists of **high scores**, **home runs**, **good players,** and a **successful team**.

We will be looking at and analyzing the win rate of teams, average scores, average hits, and other statistics in order to determine how entertaining a team is. We will then compare it to the local parks concession and ticket price costs to see if the games are worth watching per money spent.

Further Context Details

The games enjoyment will be rated with a Game Quality Score: starting pitcher quality / quality start prob, team home runs, runs, run differential, and win %

Late Quarter drawbacks:

We introduce a post-cutoff staleness penalty based on the expected duration of innings eight and later. Longer post-cutoff periods reduce the 9/9/9 optimality score, representing warming beer and cooling food.

Data Overview

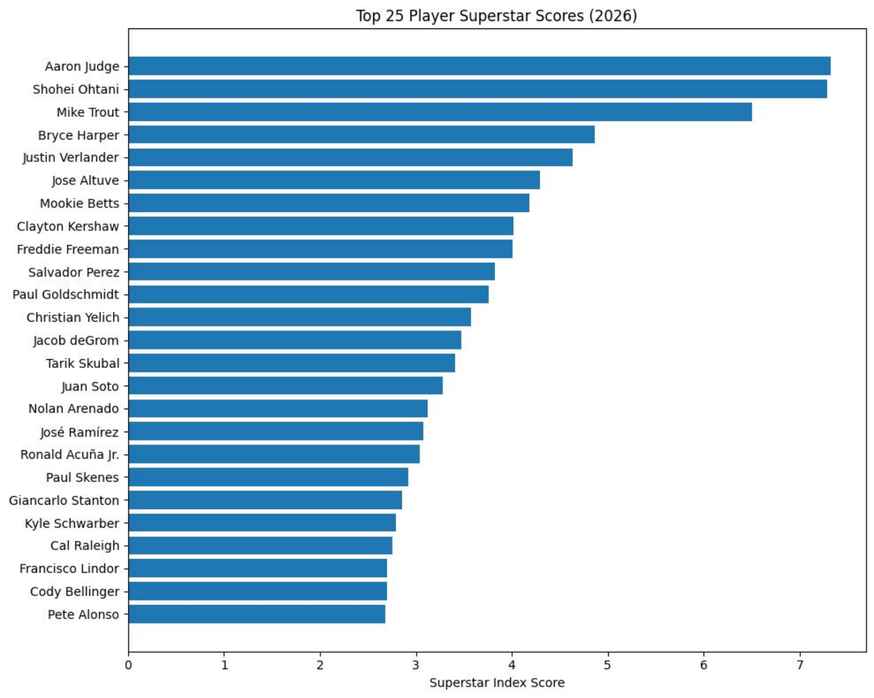
-Concession prices were selected because the 9/9/9 challenge directly depends on the cost of: 9 beers 9 hot dogs
-Ticket prices were excluded from the current model because reliable historical primary-market game-level ticket prices were not available.
- MLB StatsAPI was used for game-level and player-level data because it provides consistent data across teams and seasons.

Variable Name	Measurement
concession_999_cost	Total cost of 9 beers and 9 hot dogs
game_quality_raw	Exciting/successful the game was for that team
lineup_superstar_factor_raw	Power from actual starting lineup
current_season_leader_factor_raw	Currently near league leads in HRs, hits, strikeouts, or saves
staleness_raw	Late-game waiting time after the 7th inning
999_optimality_index	Normalized 9/9/9 score

Data Data Cleaning And Preparation

<u>Main Cleaning Decisions:</u>

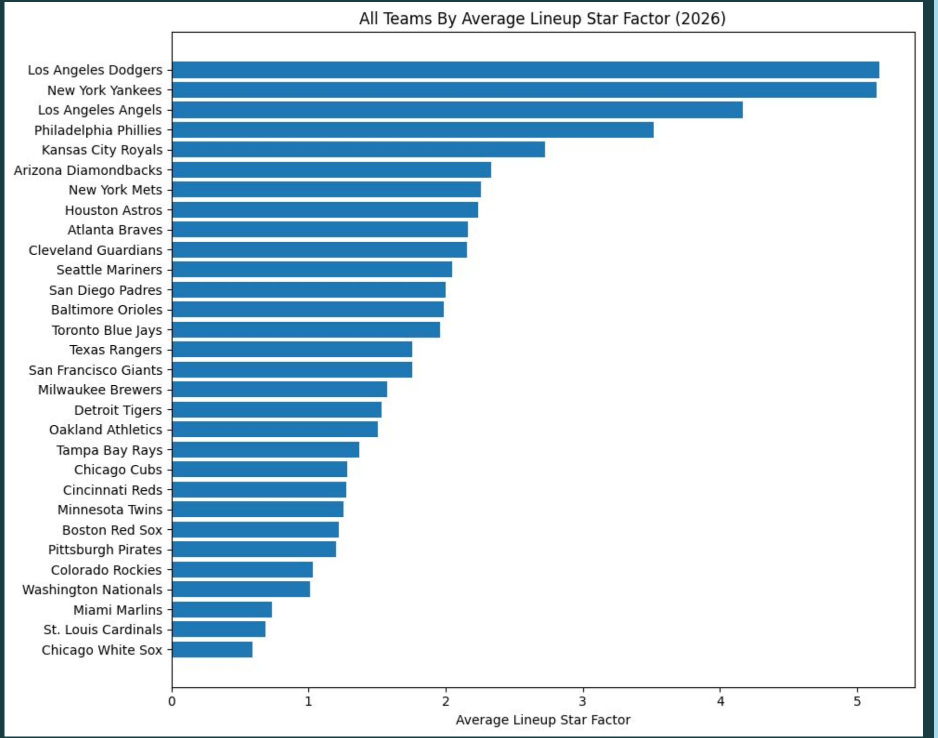
- The unit of analysis is a team-game, so each MLB game has one row per team.
- Team names were standardized across sources.
- Postponed/incomplete games were removed.
- Concession prices were merged by team-season.
- Actual starting lineups were used for the superstar factor.
- Current-season leader variables are lagged through the day before the game to avoid leakage.



Superstar Factor

Created using:
- prior-season player performance
- career awards
- actual starting lineups

Awards were weighted by importance:
- MVP and Cy Young received the highest weights
- All-Star, Gold Glove, Silver Slugger, and similar awards received smaller weights

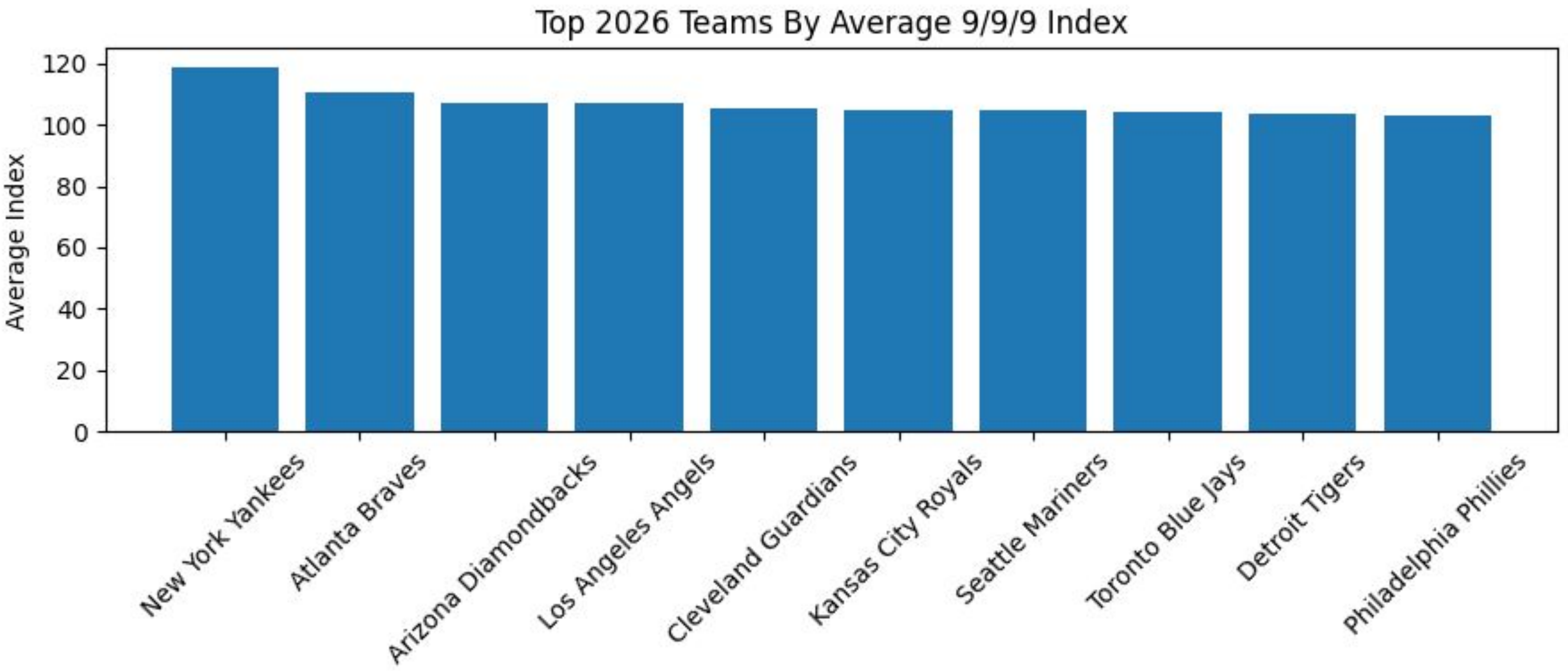


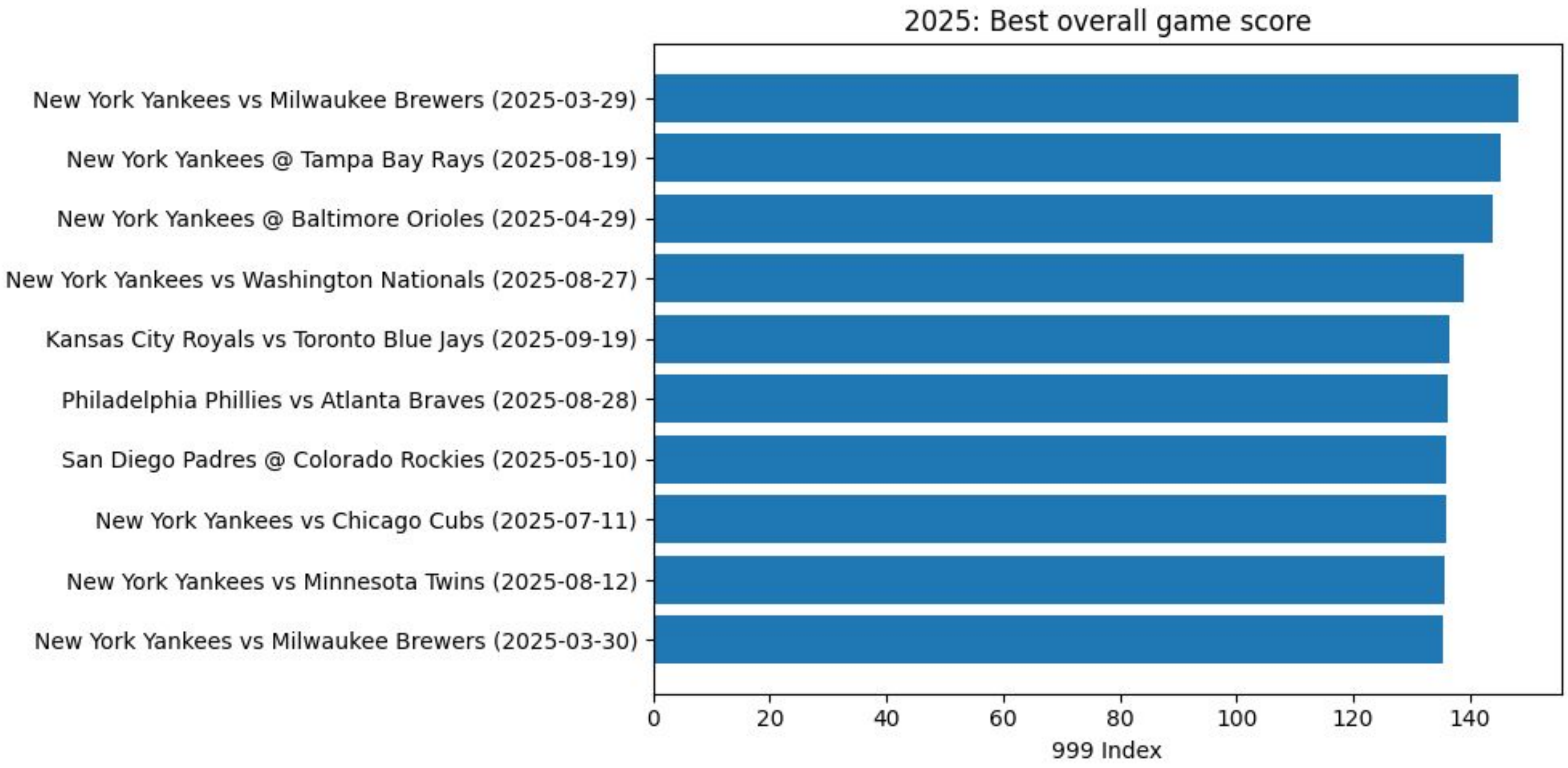
Star Factor Average

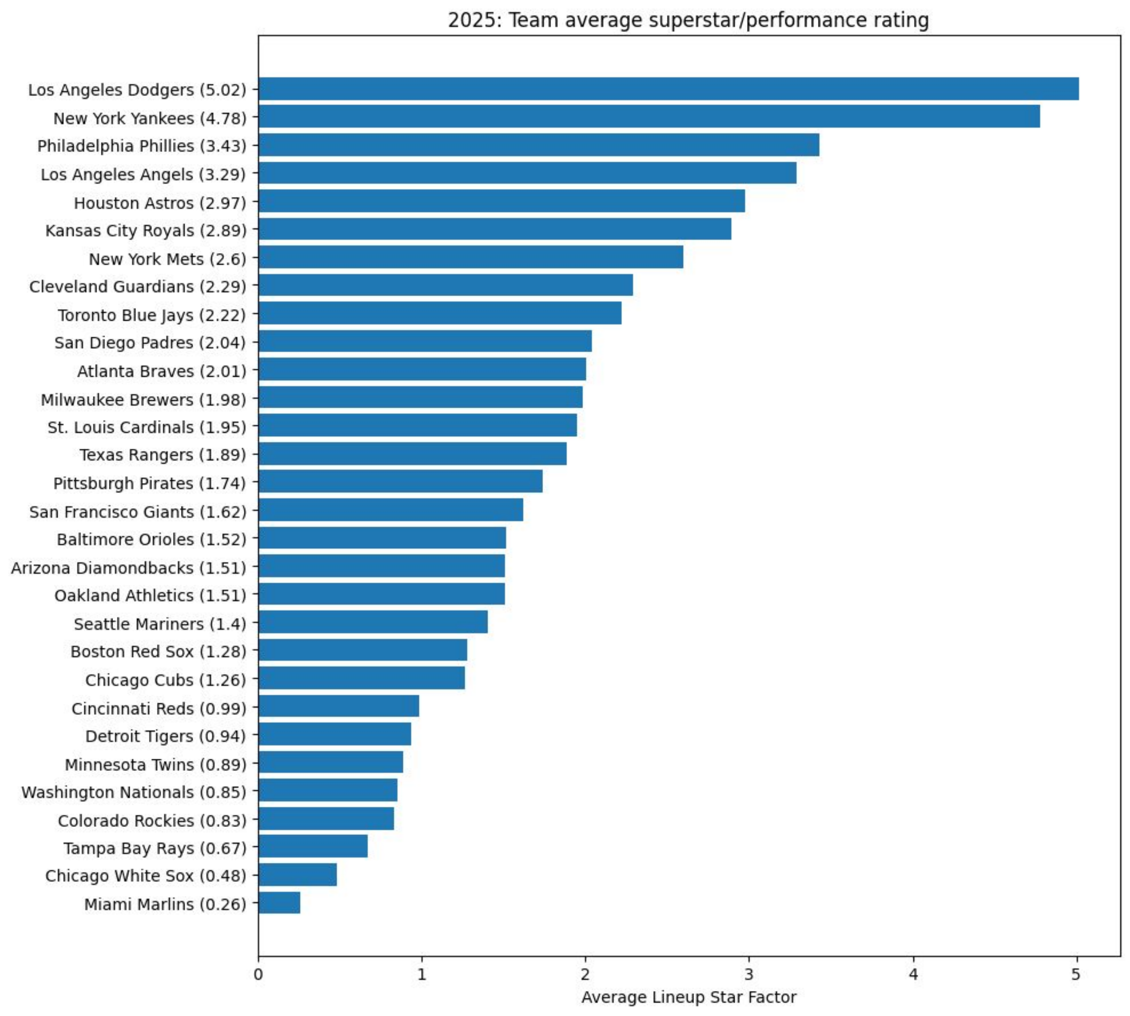
Calculated by Adding the Star Factor of each member of a team, then dividing by number of players

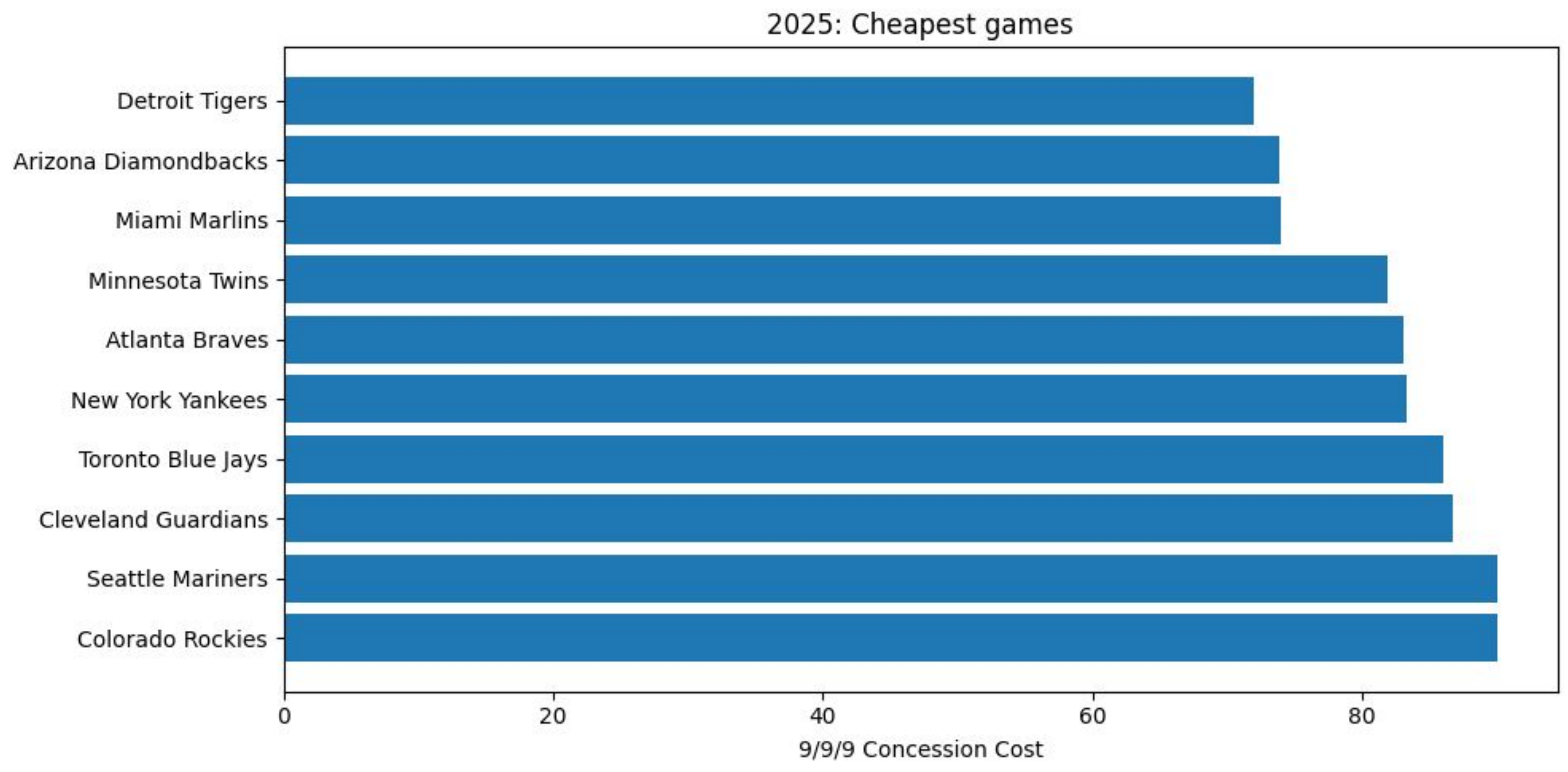
A current-season leader factor was added for players near the league lead in:

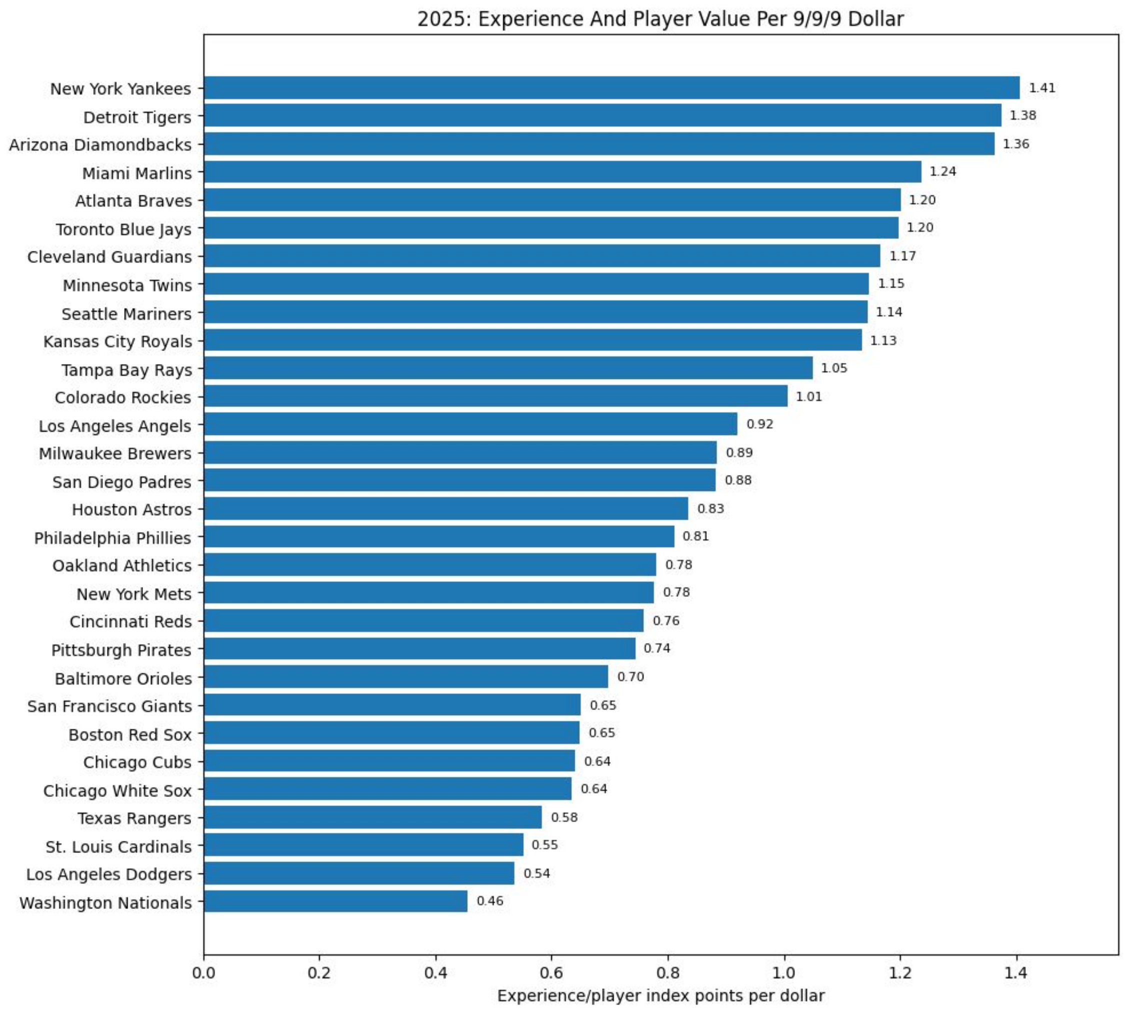
- home runs
- hits
- pitcher strikeouts
- saves


Top 2026 Teams By Average 9/9/9 Index
Average Index
0
20
40
60
80
100
120
New York Yankees
Atlanta Braves
Arizona Diamondbacks
Los Angeles Angels
Cleveland Guardians
Kansas City Royals
Seattle Mariners
Toronto Blue Jays
Detroit Tigers
Philadelphia Phillies


2025: Best overall game score
New York Yankees vs Milwaukee Brewers (2025-03-29)
New York Yankees @ Tampa Bay Rays (2025-08-19)
New York Yankees @ Baltimore Orioles (2025-04-29)
New York Yankees vs Washington Nationals (2025-08-27)
Kansas City Royals vs Toronto Blue Jays (2025-09-19)
Philadelphia Phillies vs Atlanta Braves (2025-08-28)
San Diego Padres @ Colorado Rockies (2025-05-10)
New York Yankees vs Chicago Cubs (2025-07-11)
New York Yankees vs Minnesota Twins (2025-08-12)
New York Yankees vs Milwaukee Brewers (2025-03-30)
0
20
40
60
80
100
120
140
999 Index


2025: Team average superstar/performance rating
Los Angeles Dodgers (5.02)
New York Yankees (4.78)
Philadelphia Phillies (3.43)
Los Angeles Angels (3.29)
Houston Astros (2.97)
Kansas City Royals (2.89)
New York Mets (2.6)
Cleveland Guardians (2.29)
Toronto Blue Jays (2.22)
San Diego Padres (2.04)
Atlanta Braves (2.01)
Milwaukee Brewers (1.98)
St. Louis Cardinals (1.95)
Texas Rangers (1.89)
Pittsburgh Pirates (1.74)
San Francisco Giants (1.62)
Baltimore Orioles (1.52)
Arizona Diamondbacks (1.51)
Oakland Athletics (1.51)
Seattle Mariners (1.4)
Boston Red Sox (1.28)
Chicago Cubs (1.26)
Cincinnati Reds (0.99)
Detroit Tigers (0.94)
Minnesota Twins (0.89)
Washington Nationals (0.85)
Colorado Rockies (0.83)
Tampa Bay Rays (0.67)
Chicago White Sox (0.48)
Miami Marlins (0.26)
0
1
2
3
4
5
Average Lineup Star Factor


2025: Cheapest games
Detroit Tigers
Arizona Diamondbacks
Miami Marlins
Minnesota Twins
Atlanta Braves
New York Yankees
Toronto Blue Jays
Cleveland Guardians
Seattle Mariners
Colorado Rockies
0
20
40
60
80
9/9/9 Concession Cost



2025: Experience And Player Value Per 9/9/9 Dollar
New York Yankees 1.41
Detroit Tigers 1.38
Arizona Diamondbacks 1.36
Miami Marlins 1.24
Atlanta Braves 1.20
Toronto Blue Jays 1.20
Cleveland Guardians 1.17
Minnesota Twins 1.15
Seattle Mariners 1.14
Kansas City Royals 1.13
Tampa Bay Rays 1.05
Colorado Rockies 1.01
Los Angeles Angels 0.92
Milwaukee Brewers 0.89
San Diego Padres 0.88
Houston Astros 0.83
Philadelphia Phillies 0.81
Oakland Athletics 0.78
New York Mets 0.78
Cincinnati Reds 0.76
Pittsburgh Pirates 0.74
Baltimore Orioles 0.70
San Francisco Giants 0.65
Boston Red Sox 0.65
Chicago Cubs 0.64
Chicago White Sox 0.64
Texas Rangers 0.58
St. Louis Cardinals 0.55
Los Angeles Dodgers 0.54
Washington Nationals 0.46
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Experience/player index points per dollar

Analysis of Graphs

Stand Outs

The Yankees are clear favors for the 9/9/9 Index, Overall Game Score, Average Superstar Rating, earning a high entertainment rating.

The Dodgers overall experience fall behind due to high concession pricings.

Cheap Alternative

The Detroit Tigers have the cheapest games, while also having a great 9/9/9 Index. By doing so, the Tigers have a great value per dollar ratio, and are the best budget option to watch.